ECDLP Hardness: Theorem, Bound, Heuristic, and Attack Graph

A claim-level corpus and research-obligation map

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Executive result

The assembled corpus contains **82 claims, 119 typed logical edges, 49 source records, and 12 prioritized research targets**. It separates five propositions that are often collapsed into "elliptic curves are secure":

1. the selected subgroup is large and avoids known exceptional attacks;
2. generic algorithms need birthday-order work in the declared oracle model;
3. preprocessing, memory, parallelism, and multiple targets are charged consistently;
4. no known coordinate-aware classical algorithm beats the best generic method on the named family;

5. the adversary is classical and the implementation exposes only valid subgroup operations.

Only parts 1–3 have strong theorem support. Part 4 remains an empirical and literature-backed assumption for standard prime-field curves; part 5 is a system model. There is no known unconditional superpolynomial classical lower bound for ECDLP on a named standard curve in the unrestricted computation model.

The graph does identify a disciplined way to "go after" the assumption. A classical sub-rho attack must exploit at least one capability excluded from the generic lower bounds: coordinates/algebra, special curve structure, target-independent setup under a different accounting model, or a stronger oracle such as a cheap endomorphism. The most credible active branch is coordinate algebra, but it does not become an attack until factor-base generation, relation yield, polynomial solving, rank, sparse linear algebra, individual descent, memory, and verification all fit one sub-rho cost bound.

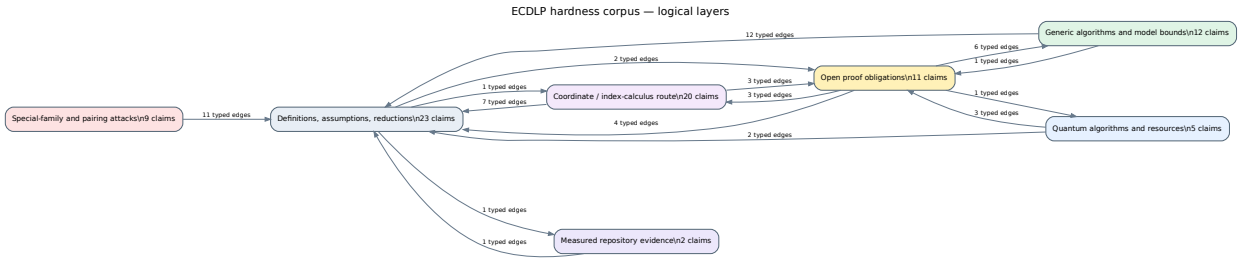


Figure 1: Seven-layer overview of the corpus. The full 82-node graph is in `graph.svg`.

Scope and terminology

The corpus covers classical and quantum ECDLP, CDH/DDH reduction boundaries, generic-group lower bounds, preprocessing, conventional algorithms, exceptional-curve attacks, pairing transfers, Weil descent, summation-polynomial index calculus, prime-field algebraic proposals, isogeny transport, quantum resources, and the first protocol/implementation boundary layer.

The phrase in the request may refer to either of two different techniques, so both are included:

- **index calculus:** factor base, relation collection, sparse linear algebra, and descent;
- **xedni calculus:** Silverman's "index backwards" lift to a rational curve followed by a Mordell–Weil dependence search.

Xedni is not a synonym for Semaev index calculus. The two branches have different mechanisms and different obstructions.

This report is a research map, not a security proof for a product and not a claim of a new ECDLP break. Exact records are in `corpus.json`, `corpus-extension.json`, `claims.tsv`, and `sources.tsv`.

Load-bearing theorems and bounds

Claim	Exact form retained in the graph	Hypotheses / model	What it does not prove
Hasse	`#E(F_q)=q+1-t`, `|t|<=2 sqrt(q)`	elliptic curve over a finite field	prime order or hardness
Pohlig–Hellman	reduce DLP modulo the group order to prime-power components	known group order/factorization	hardness of the largest prime component
BSGS	`O(sqrt(r))` group operations and storage	generic cyclic group	a memory-light optimum

Claim	Exact form retained in the graph	Hypotheses / model	What it does not prove
Pollard rho	expected `O(sqrt(r))`; random-mapping repeat constant about `sqrt(pi r/2)`	heuristic random walk; useful collision and restart conditions	an unconditional wall-time constant
Shoup 1997	generic DLP success `O(m^2/p)` after `m` oracle queries; constant success needs `Omega(sqrt(p))`	random encoding; `p` largest prime divisor; generic oracle	coordinate algorithms, pairings, polynomial systems, cheap extra maps
CGK 2018 advice	`S T^2 = soft-Omega(epsilon N)`	prime-order generic group; `S` advice bits; `T` online queries; success `epsilon`	uncharged or coordinate-augmented advice outside the interface
CGK 2018 setup	`P T + T^2 = Omega(epsilon N)`	paper's generic preprocessing experiment	a many-user conversion without matching experiments
Smart trace-one	polynomial/linear-basic-operation attack when `#E(F_p)=p`	anomalous prime-field curve; p-adic lift	ordinary non-anomalous curves
MOV / Frey–Rueck	transfer order-`r` ECDLP to a multiplicative group over an embedding field	nondegenerate pairing; suitable extension	that small `k` alone determines cost
GHS	conditional `O(q^(2+epsilon))` on specified characteristic-two families with fixed composite extension degree	transfer kernel does not kill the large subgroup; paper's family	arbitrary binary or prime-field curves
Gaudry fixed extension	quoted `soft-O(q^(2-2/n))` for fixed `n>=3` under PDP hypotheses	extension field; decomposition/solver/rank assumptions; hidden exponential dependence on `n`	a prime-field result obtained by putting `n=1`
Diem 2011	expected `exp(O(max(log q,n^2)))` in the paper's input model	specified extension-field sequence and randomized construction	a uniform subexponential attack on named prime curves
Shor	polynomial-time DLP in finite abelian groups	scalable fault-tolerant quantum computation	practical physical resources on one architecture
JKSST xedni analysis	one-shot `<= C0/p`; uniform conditional hazard gives `E[T]=Omega(p)`	selected-lift height/Lang; reduction, usefulness, bounded cofactor, adaptive recurrence	an unconditional lower bound on every lifting algorithm

The logical asymmetry is essential. Generic lower bounds are rigorous but narrow; unrestricted classical hardness is broad but assumed.

Assumption lattice

DLP, CDH, and DDH are not interchangeable

An ECDLP oracle solves CDH, so `CDH <= DLP`. A CDH oracle decides DDH, so `DDH <= CDH`. Therefore:

- CDH hardness implies DLP hardness;

- DDH hardness implies CDH and DLP hardness;
- DLP hardness alone does **not** imply CDH hardness;
- CDH hardness alone does **not** imply DDH hardness.

Pairings make the distinction concrete: a source group can have easy DDH while CDH is still assumed hard (a gap-DH setting). Protocols must therefore be attached to the assumption their proof actually uses, not labeled merely "ECDLP-based."

Parameter screens are necessary, not sufficient

NIST SP 800-186 makes the conventional exclusions explicit: a large prime-order base point, small cofactor, non-anomalous order, and large embedding degree. Twist orders and public-point validation matter at the system boundary. Passing this checklist removes named attacks; it does not prove unrestricted ECDLP hardness.

The security parameter is the prime subgroup order `r`, not simply the field size. Under the generic rho estimate, a random full-width subgroup of approximately `2^256` offers roughly a `2^128` operation exponent before constants and multi-target effects. This is an upper-cost estimate for known attacks, not a lower-bound theorem in the unrestricted model.

Generic algorithms and preprocessing

Upper and lower bounds nearly meet inside the model

BSGS and rho supply `O(sqrt(r))` generic upper bounds; Shoup supplies the matching `Omega(sqrt(r))` query scale for constant-success generic attacks on a prime-order subgroup. The result explains why a purely black-box rearrangement can improve constants, memory, or parallelism but not the square-root exponent.

van Oorschot–Wiener distinguished-point search gives near-linear wall-clock parallelism while total collision work remains birthday order. In its notation, for state space `N`, useful-collision probability `u`, `m` processors, distinguished density `theta`, and iteration time `t`, the detection-time model is

`T = (sqrt(pi N/(2u))/m + 1/theta) t`.

That expression carries assumptions: a shared random-looking mapping, controlled endpoint storage/communication, and a useful collision. It is not a universal hardware constant.

The first concrete audit result

`GENERIC_INTERFACE_AUDIT.md` traces every current decomposition oracle into source. The audit uses four interfaces:

- **G0:** standard opaque generic group;
- **G1:** G0 plus fully charged target-independent advice/preprocessing;
- **GA:** G0 plus a cheap known-scalar automorphism;
- **C:** coordinate and algebraic access.

The initial conclusions are:

- subtract, MITM, k-sum, rootfind-MITM, and pair enumeration are generic table algorithms once their base is supplied;
- the roots of specialized `S3(x_R,x_F,X)` are exactly `x(R-F)` and `x(R+F)`, so arity-two root finding does not create a new algebraic search space;
- coordinate batching can change the implementation constant, and smallest-`x` factor-base generation remains representation-specific;
- Koblitz/GLV Frobenius folding needs GA for a same-cost claim: a generic simulator can compute `[lambda]P`, but not for the price of one native Frobenius map;
- F4/F5/SAT, PKM tower, Weil descent, and lattice small-root solvers remain in C and outside Shoup/CGK.

This closes a bookkeeping ambiguity without closing the actual algebraic attack surface.

Index calculus as an obligation graph

End-to-end cost identity

For factor-base size `B`, arity `m`, subgroup order `r`, and one decomposition attempt cost `C_PDP`, a useful first-order ledger is

`T_total = P_setup + R_trials C_PDP + T_filter + T_LA + T_descent + T_verify`.

The mean ordered decomposition multiplicity over targets is exactly `B^m/r` by double counting. Approximating target success by

`p_dec ~= 1-exp(-B^m/r)`

adds a Poisson/independence heuristic. In the sparse regime, collecting about `B` relations costs roughly

`R_trials ~= B / p_dec ~= r / B^(m-1)`.

If `B=r^beta`, the relation-trial exponent is `1-(m-1)beta`. Conventional sparse linear algebra contributes about `B^2`, exponent `2 beta`, before block/rank constants. Ignoring decomposition-solver cost and balancing these terms gives

`beta=1/(m+1)`, total exponent `2/(m+1)`.

This tempting algebra is exactly where unjustified claims arise. For `m>3` it appears sub-rho only because `C_PDP` has been treated as free. A generic enumeration/table oracle pays back that apparent gain through setup, memory, or per-target work; Shoup forbids an end-to-end generic contradiction. A coordinate solver can escape Shoup, but then its solving degree, matrix dimensions, rational roots, and memory must be bounded.

Independent obligations

The graph isolates the following nodes instead of hiding them inside one "index calculus heuristic":

1. factor-base cardinality and construction cost;
2. algebraic-closure solution count and rational-point yield;
3. target-to-target variance and exceptional specializations;
4. zero-dimensionality of the descended PDP system;
5. actual Groebner solving degree, not merely first fall;
6. Macaulay matrix dimensions, sparsity, and peak memory;
7. relation independence and full factor-base rank;
8. sparse linear algebra and filtering;
9. individual logarithm/descent;
10. setup reuse, advice bits, and online success;
11. final verification in the original group.

The live prime-field route is PKM/tower or another genuinely algebraic factor-base predicate. Small measured degree plateaus are useful falsifiers and engineering data, but do not establish bounded degree at cryptographic scale.

Extension and binary fields

Gaudry/Diem-style gains are real on specified extension-field regimes. They are not evidence for a prime-field attack. Over binary fields, summation-polynomial Weil descent produces Boolean/multivariate systems whose cost depends on actual first/last-fall and solving degree. GHS is a different descent—to a curve/Jacobian—and depends strongly on the resulting genus and transfer kernel.

For the repository's degree-131 Koblitz target, existing structural work severely limits Frobenius-stable linear subspaces and makes the classified GHS genus enormous. These are scoped negatives; nonlinear or non-invariant algebraic bases and unresolved m=4 solver behavior remain separate questions.

Xedni calculus

Silverman's xedni proposal takes several random combinations of `P` and `Q`, lifts them to rational points on an elliptic curve over `Q`, and hopes for a Mordell–Weil linear dependence. Reducing such a dependence modulo `p` gives the desired scalar relation.

Jacobson, Koblitz, Silverman, Stein, and Teske analyze this mechanism directly. Their displayed `C0/p` success bound and linear waiting-time obstruction depend on a chain: selected-point height versus a global minimal discriminant, Lang's height gap, Mordell–Weil packing, good reduction and subgroup sampling, selection of a relation with nonzero denominator, a bounded-cofactor conversion from subgroup order `q` to field prime `p`, and a history-uniform hazard statement for repeated adaptive trials. Under that complete chain the runtime is `Omega(p)`, asymptotically worse than rho. Their experiments also find suitable lifts prohibitively unlikely at practical sizes.

A 2026 kernel-checked Lean audit proves the finite core without erasing those bridges: exact coefficient-box counts, a packing theorem with fiber multiplicity, cyclic-log extraction, fixed-direction useful-event probability `(q-1)/q^2`, the Hasse/cofactor conversion, and the conditional hazard-to-`Omega(p)` theorem. It also gives explicit counterexamples to several overbroad readings. Passing the proof gate establishes these endpoint theorems, not their correspondence to the actual lift algorithm.

The repository studied a different, frozen function-field lifting family. For that exact family it derives

`P_lift(p) = (1-2/p+p^-3)/(2p^3) = Theta(p^-3)`.

The equality is exact within its family, not a universal theorem about every lifting construction. Together the results strongly disfavor the historical and frozen routes, but the graph keeps the end-to-end JKSST bridge as an explicit open obligation rather than calling every conceivable lifting strategy unconditionally closed.

Special-family and transfer attacks

The graph treats known attacks as conditions, not folklore exceptions:

- **Pohlig–Hellman:** small prime factors are easy; price the largest prime subgroup.
- **Anomalous curves:** `#E(F_p)=p` admits additive/p-adic transfer attacks.
- **MOV/Frey–Rueck:** low embedding degree transfers ECDLP to finite-field DLP.
- **Supersingular curves:** have very small embedding degree in the classical ECC setting.
- **GHS:** can beat generic ECDLP on specific composite binary extensions, subject to its kernel and genus conditions.
- **Cheon:** auxiliary powers can lower cost under divisibility conditions; this is an augmented-input problem, not plain ECDLP.
- **Invalid-curve/small-subgroup attacks:** leak implementation scalars by leaving the intended group; they do not solve valid-input ECDLP.
- **Isogeny transport:** random reducibility within an endomorphism-ring scope does not provide a free coordinate-aware factor-base transport across arbitrary levels.

Quantum boundary

Shor removes asymptotic ECDLP hardness on scalable fault-tolerant quantum computers. Concrete estimates form a moving, architecture-dependent ladder:

- Roetteler et al. (2017) give one logical construction bounded by `9n+2 ceil(log2 n)+10` logical qubits and `448 n^3 log2(n)+4090 n^3` Toffoli gates.
- Haener et al. (2020) improve arithmetic with windowing, pebbling, and inversion changes, emphasizing width/depth/T-count tradeoffs.
- Gouzien et al. (2023) estimate 9 hours and 126133 cat qubits for a 256-bit ECDLP under a specific 2D repetition-cat architecture, `10^-5` loss ratio, and 500 ns cycle.

- A September 2026 preprint reports a secp256k1 circuit of about 1450 logical qubits and 40 million Toffolis, compiled to 19397 physical trapped-ion qubits, 25.7 days, and estimated 63% success on its proposed architecture.

These are not mutually contradictory security levels. They optimize different circuits and assume different error correction, routing, factories, clocks, and success budgets. The last item is retained as a primary-read preprint, not treated as independently replicated.

Research priorities

Priority	Target	Acceptance criterion in brief
1	Concrete generic-interface audit	every live method classified by setup, online operations, advice, and exact escaping predicate
2	Algebraic solution-count falsifier	exact/certified counts for structured and random controls under one convention
2	End-to-end prime-field IC ledger	one symbolic maximum over setup, relations, solver, rank, LA, descent, memory, and verification
3	PKM solving-degree growth	uncensored larger rungs plus checked Hilbert-series prediction
4	Binary m=4 degree ladder	matched larger cells and shared cost-unit conversion
5	Coordinate transport theorem	explicit interface and proof or counterexample after the withdrawn overclaim
5	Xedni bridge-composition audit	instantiate selected lifts, height, reduction, usefulness, cofactor, and adaptive hazards for one explicit family
6	Automorphism preprocessing model	GA oracle theorem or cost-preserving reduction with orbit/stabilizer accounting
7	Named-curve quantum ladder	verified circuits and architecture/error assumptions
8	Multi-target/nonuniform model	target distribution and all advice/setup units matched to CGK
9	Pairing target-field census	exact `q,r,k`, target representation, and finite-field DLP cost for named curves
10	Protocol assumption graph	proof notion, exact assumption, model, validation rule, and reduction loss per protocol

The best immediate theoretical target is the automorphism/preprocessing bridge; the best attack-oriented target is the end-to-end prime-field algebraic ledger plus the solution-count falsifier. The former can close misuse of existing theorems. The latter can reveal whether a proposed coordinate base changes recognizability/solver cost or only rearranges generic yield.

Evidence and limitations

Every externally proved claim must cite at least one `primary_read` source under the validator. Every primary-read record carries a SHA-256. Inaccessible primary papers are marked metadata/secondary and are not used as sole evidence for an external theorem. Internal formal/analytic claims remain `proved_internal_unreviewed`.

The literature search is deep and branch-complete for load-bearing claims, but it is not an assertion that all ECDLP publications have been read. The most important remaining primary verification targets are original MOV/Frey–Rueck, Gaudry/Joux–Vitse, and protocol-specific proof papers. `LITERATURE_SEARCH.md` records those limits and the discovery ledger.

The full graph is necessarily dense. `overview.svg` is the readable layer map; `graph.svg` is the claim-level audit artifact; `claims.tsv` and the JSON corpus are the preferred exact interfaces.

Conclusion

ECDLP security is not one theorem. It is a stack:

`valid subgroup and curve screens`

`+ generic birthday lower bounds in explicit models`

`+ no verified special-family or coordinate-aware faster attack`

`+ protocol reduction and implementation validation`

`+ a classical-adversary qualifier`.

The rigorous generic results are strong enough to tell us where not to spend time: relabeling group subtraction as root finding, building larger generic sum tables, or claiming a free preprocessing/automorphism quotient cannot by itself establish a sub-rho exponent. The remaining credible research surface is narrower and sharper: coordinate predicates whose algebraic solution cost scales favorably, exact transport interfaces, or a new model-changing mechanism. The corpus turns each of those into a falsifiable obligation rather than an undifferentiated hope.

Selected sources

- Victor Shoup, Lower Bounds for Discrete Logarithms and Related Problems, EUROCRYPT 1997.
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- Peter Shor, Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms, SIAM Journal on Computing 1997.
- NIST, SP 800-186: Elliptic Curve Domain Parameters, 2023.
- Martin Roetteler et al., Quantum Resource Estimates for Computing ECDLP, 2017.
- Thomas Haener et al., Improved Quantum Circuits for ECDLP, 2020.
- Elie Gouzien et al., Performance Analysis of a Repetition Cat Code Architecture, 2023.

The complete source ledger, including hashes and verification status, is `sources.tsv`.